

REFNO: 074896

Roll No. _____

B.Sc. (Hons.) SEMESTER II EXAMINATION 2023-24
MATHEMATICS
MTB - 201 : Calculus - II

Time : Three hours

Max. Marks : 70

(WRITE YOUR ROLL NO. AT THE TOP IMMEDIATELY ON THE RECEIPT OF THIS QUESTION PAPER)

Instructions: Solve any five questions. Question 1 is compulsory.

1. (a) Let f be a function defined by $f(x, y) = \frac{xy^2}{x^2+y^4}$ for all $(x, y) \in \mathbb{R}^2$ with $(x, y) \neq (0, 0)$. Check that, whether the limit $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists or not. **2**

- (b) Let f be a function defined by

$$f(x, y) = \begin{cases} \frac{x^3-y^3}{x^2+y^2}, & x \neq y \\ 0, & x = y \end{cases}$$

Find the first partial derivatives f_x and f_y at the origin and test the continuity of f_x at the origin. **2**

- (c) If $z = \sin(x^2 + y^2)$, where $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then find $\frac{dz}{dx}$. **2**
- (d) Define a homogeneous function of degree n in the 4 variables x, y, z, w and give an example. **2**
- (e) Let U be a nonempty open set in $\mathbb{R}^2$ and $f : U \rightarrow \mathbb{R}$ a function. Give definition of derivative of f at $(a, b) \in U$. Find the derivative of the function f defined by $f(x, y) = xy^2 + xy^3$, $(x, y) \in (-2, 2) \times (-2, 2)$ at $(-1, -1)$. **2**
- (f) Change the order of integration in integral $\int_0^1 dy \int_y^1 e^{x^2} dx$ and hence evaluate it. **2**
- (g) Find the volume of the solid, which is bounded by the cylinder $x^2 + y^2 = 1$ and the planes $y + z = 1$ and $z = 0$. **2**
2. (a) Let D be a nonempty set in $\mathbb{R}^2$ and $c = (a, b) \in \mathbb{R}^2$ such that $B_r(c) \setminus \{c\} \subseteq D$ for some $r > 0$. Let $f, g : D \rightarrow \mathbb{R}$ be functions such that $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = \ell$ and $\lim_{(x,y) \rightarrow (a,b)} g(x, y) = m$. Show that $\lim_{(x,y) \rightarrow (a,b)} f(x, y)g(x, y) = \ell m$. **5**
- (b) State and prove Young's theorem for a function two variables. Verify Young's theorem by an example. **9**
3. (a) Let D be a nonempty open set in $\mathbb{R}^2$ and $f : D \rightarrow \mathbb{R}$ a function such that f is differentiable at $(a, b) \in D$. Show that f is continuous at (a, b) . Give an example of a function $f(x, y)$ such that f is continuous at $(0, 0)$, but it is not differentiable at $(0, 0)$. **6**
- (b) Let C be a nonempty subset of $\mathbb{R}^2$ with $(a, b) \in \text{int}(C)$ and let $f : C \rightarrow \mathbb{R}$ be a function. Suppose there exists $r > 0$ such that
- (i) $B_r((a, b)) \subseteq C$,
 - (ii) $f_x(x, y)$ exists for all $(x, y) \in B_r((a, b))$ and f_x is continuous at (a, b) ,
 - (iii) f_y exists at (a, b) .
- Show that f is differentiable at (a, b) . **8**

(2)

4. (a) State and prove Euler's theorem for homogeneous functions of degree n in the variables x, y and z . Verify Euler's theorem for a function $f(x, y, z)$ of degree 3. **5**
- (b) Prove that $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ is invariant for change of rectangular axes. **5**
- (c) Show that the functions $u = x + 2y + 2z$, $v = 3x + 4y - 2z$, $w = 11x + 18y - 2z$ are not independent and find the relation among them. **4**
5. (a) State and prove Taylor's theorem for expansion of function f of two variables about the point (a, b) . **7**
- (b) Find the minimum value of $x^2 + y^2 + z^2$ subject to the condition $xy + yz + zx = 12$. **7**
6. (a) Evaluate the integral $\iint (x+y)^2 dx dy$ over the area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. **7**
- (b) Evaluate the integral $\iint r \sin \theta dr d\theta$ over the area of the cardioid $r = a(1 + \cos \theta)$ above the initial line. **7**
7. (a) Establish the relation between Beta and Gamma functions. **7**
- (b) Evaluate the integral

$$\iiint x^{l-1} y^{m-1} z^{n-1} (1 - x - y - z) dx dy dz$$

over the tetrahedron formed by the plane $x + y + z = 1$ and the coordinate planes. **7**

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